

Project Title: Modeling Insurance Claim Risk Using Neural Networks

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This project investigates whether machine learning and neural networks can be used to **predict insurance claim risk** using the Porto Seguro Safe Driver dataset from Kaggle. The goal is to determine whether a policyholder will file an auto insurance claim in the following year, formulated as a binary classification problem. The dataset contains over **595,000 insurance policies** with **59 fully anonymized features**, making it a realistic example of large-scale, real-world tabular data with limited domain interpretability.

We selected this dataset because it introduces several practical challenges common in applied AI: **mixed feature types**(continuous, binary, and categorical), **non-standard missing value encoding** (-1 instead of NaN), and **severe class imbalance**, with only about 3–4% of policies associated with an actual claim. These characteristics make standard accuracy unreliable as an evaluation metric and motivate the use of **AUC** for model comparison. Our project demonstrates an **end-to-end modeling pipeline**. We begin with careful preprocessing by separating features by type, analyzing missing value patterns, and removing extremely sparse variables. Continuous features are median-imputed and standardized, binary features preserve their original semantics, and categorical features retain missing values as an explicit “unknown” category before one-hot encoding. The result is a fully numeric feature matrix suitable for both classical machine learning models and neural networks.

Using this pipeline, we evaluate **logistic regression and a decision tree** as interpretable baseline models, followed by a **multilayer perceptron (MLP)** to explore whether neural networks can capture additional nonlinear structure. While the neural network achieves a modest improvement in AUC over the baselines, the results also highlight key limitations, including sensitivity to learning rate, training instability, and diminishing returns from deeper architectures.

Overall, this project shows how neural networks can be applied to anonymized insurance data, while also emphasizing their trade-offs relative to simpler models. The working example illustrates data preprocessing, model comparison, and metric selection in an imbalanced, real-world classification problem, helping readers understand both the practical benefits and limitations of AI on tabular data.